How-To Library

Informatica

Configuring Git Version Control for Model Repository Service in 10.2 HotFix 1

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Abstract

You can integrate a Model repository with a Perforce, Subversion, or Git version control system. This article discusses how to integrate a Git system with a Model Repository Service in 10.2 HotFix 1.

Supported Versions

- Big Data Management 10.2 HotFix 1
- Big Data Quality 10.2 HotFix 1
- Data Quality 10.2 HotFix 1

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Overview

You can integrate a Model Repository Service with a version control system, such as Perforce, Subversion, or Git system. Git is a distributed version control system.

A version control system allows one user at a time to check out, edit, and save an object. When you save an object, the object is saved in the Model repository. After you check in the object, a version is created in the version control system. The version control system maintains a history of all the versions. You can edit only the latest version of the object. You can view the other versions of the object in read-only mode. You can roll back to a previous version or reassign the checked-out state of objects to another user. A version control system protects a Model repository object from unwanted updates because it does not allow multiple users to edit an object at once.

To integrate the Model Repository Service with Git version control system, you can configure the versioning properties of the Model Repository Service. Then, set the **git_local_repository_path** custom property with the file path of the local Git repository, and recycle the Model Repository Service.

When you choose the Git version control system, you can configure the following components:

Remote Git repository

You need access to the remote repository on the Git server. To configure the version control system, you need the URL, user name, and password of the remote repository. You can use HTTP or HTTPS protocol to access the remote Git repository.

Local Git repository

Create a directory on the machine that hosts the Model Repository Service to serve as the local Git repository.

The directory must meet the following requirements:

- Access to all the client machines.
- Access to the backup nodes for the Model Repository Service after you enable high availability.
- Support for NFS, FAT32, and NTFS file systems.
- Have a unique name.
- Have read, write, and execute permissions.

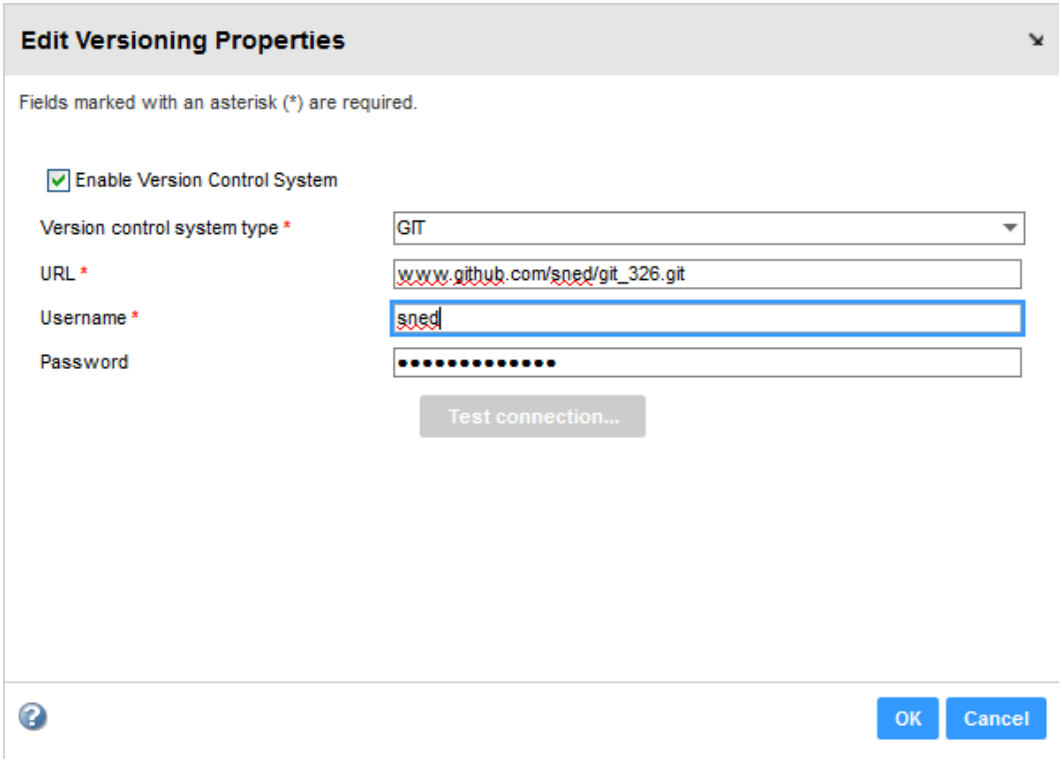
When you check in an object, the Git version control system checks in the object. It saves a copy in the remote Git repository and local Git repository. If the remote Git repository is inaccessible or if it shuts down unexpectedly, you can access the local Git repository to view all the versions and edit the latest version of the object.

Configuring a Model Repository Service for Git Version Control System

In an Informatica domain, you can configure the Git version control system for only one Model Repository Service.

1. In Informatica Administrator, navigate to the **Manage** > **Services and Nodes** > **Domain Navigator** section.
2. In the **Domain Navigator** section, select the Model Repository Service.
3. In the **Properties** > **Versioning Properties** section, click the **Edit** icon.
4. In the **Edit Versioning Properties** dialog box, set the following options:
   - Select the **Enable Version Control System** option.
   - In the **Version control system type** field, choose **GIT**.
   - In the **URL** field, enter the URL for the remote Git repository.
   - In the **Username** field, enter the user name for the remote Git repository.
   - In the **Password** field, enter the password of the remote Git repository user.
   - To test the connection, click **Test Connection**.

   The following image shows a sample **Edit Versioning Properties** dialog box:



5. In the **Edit Versioning Properties** dialog box, click **OK**.

   The configured properties appears in the **Versioning Properties** section.
6. In the **Properties** > **Custom Properties** section, click **Edit**.
7. In the **Edit Custom Properties** dialog box, click **New**.

8. In the **New Custom Property**, enter the following details:
   - In the **Name** field, enter **git_local_repository_path**.
   - In the **Value** field, enter the file path of the local Git repository.
9. In the **New Custom Property** dialog box, click **OK**.
10. In the **Edit Custom Properties** dialog box, click **OK**.
11. Recycle the Model repository.

    Before you recycle the service, ask Model repository users to close all editable objects. When the Model repository restarts, it checks whether the version control system is in use by another Model repository. If the version control system is connected to a different Model repository, the Model Repository Service cannot restart. You must configure the versioning properties to connect to a version control system that has not been integrated with another Model repository.
12. To synchronize the Model repository contents to the version control system, perform one of the following tasks:
    - In Informatica Administrator, select the Model repository, and click **Actions** > **Synchronize With Version Control System**, and click **OK**.
    - Run the infacmd mrs PopulateVCS command.

    When you synchronize the content, the Model repository populates the local Git repository and remote Git repository with Model repository objects. After the synchronization is complete, versioning is active for Model repository objects. All Model repository objects are checked in to the version control system. Users can check out, check in, view version history, and restore historical versions of objects.

Additional Configuration for Git Version Control on AIX Operating System

To use the Git version control system in the AIX operating system, you need to configure the Model Repository Service for TLSv1.2.

To configure the Model Repository Service for TLSv1.2, set the following parameters:

-Dhttps.protocols environment variable

Before you start the domain, you must configure the environment variable.

To configure the variable, perform the following steps:

1. In Informatica Administrator, navigate to the **Model Repository Service** > **Processes** > **Environment Variables** section.
2. In the **New Environment Variable** dialog box, enter **ACJVMCommandLineOptions** in the **Name** field, and **-Dhttps.protocols=TLSv1.2** in the **Value** field.
3. Click **OK**.

After you configure the environment variable, shut down all the services and restart the domain.

JVM option

1. In Informatica Administrator, navigate to the **Model Repository Service** > **Advanced Properties** section.
2. Append **-Dhttps.protocols=TLSv1.2** to the **JVM Command Line Options** field.

Optionally, you can run the **infacmd mrs UpdateServiceProcessOptions** command to add the **-Dhttps.protocols=TLSv1.2** option to the JVM options.

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